

ESTIMATING THE COVARIANCE DENSITY FROM OBSERVATIONS ON A FILTERED POINT PROCESS

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SUMMARY

This paper discusses methods of estimating the covariance density of a stationary point process in cases where the point process itself cannot be observed, but where observations on a filtered point process are available. Two cases are considered: the element function is either fixed or random. Typical results obtained in Monte Carlo experiments are discussed.

1. Filtered point processes

A filtered point process is a stochastic process

$$(1) \quad Y(\omega, t) = \int_{-\infty}^{\infty} w(t-u) dN(\omega, u).$$

It is assumed that $w(t)$, the element function, is quadratically integrable and zero for negative t . $N(\omega, u)$ is the counting process of a stationary point process $N(\omega, B)$, $B \in \mathcal{B}$, defined on a probability space $(\Omega, \mathcal{A}, P)$. (For definitions and notation see Steinijans (1970).) In particular $T_i(\omega)$, or simply T_i , is used to denote the instants at which the points occur and $\gamma(\tau)$ to denote the covariance density.

A method to estimate $\gamma(\tau)$ from observations $Y(t)$, $0 \leq t \leq T$, based on the covariance function of $Y(t)$, is given in Linhart (1973). We shall modify this result for the case in which the observations on $Y(t)$ are at discrete "times", i.e. at $i\tau$: $i=1, \dots, N$; $\tau > 0$. We shall also discuss the case in which the element function $w(t-T_i; X_i)$ is a random function whose distributions are determined by random vectors $X_i = (X_{i1}, \dots, X_{in_i})$. The vectors X_i are assumed to be independently and identically distributed, independent of the point process. A special case of this problem was treated in Linhart (1974).

2. Fixed element function

For convenience we write Y_i instead of $Y(i\tau)$: $i=1, \dots, N$. An estimator of the covariance density, which is simply the discrete analogue of that given in Linhart (1973),

is given by

$$(2) \quad \hat{\gamma}(s\tau) = (\hat{C}_Y(s\tau) - \hat{\nu} r(s\tau))/\alpha; s = 0, \pm 1, \pm 2, \dots$$

$$\text{where } r(v) = \int_{-\infty}^{\infty} w(t)w(t+v)dt, \alpha = \int_{-\infty}^{\infty} r(v)dv,$$

$\hat{\nu} = \frac{1}{n} \sum_{i=1}^n Y_i / N\sqrt{\alpha}$ is an estimator of the intensity of the point process and

$$\hat{C}_Y(s\tau) = \frac{1}{N-|s|} \sum_{i=1}^{N-|s|} (Y_i - \bar{Y})(Y_{i+|s|} - \bar{Y}); s=0, \pm 1, \pm 2, \dots$$

is an estimator of the covariance function of $Y(t)$. It can be shown that $\hat{\gamma}(s\tau)$ is an approximately unbiased estimator of a weighted average of $\gamma(u)$, namely

$$\int_{-\infty}^{\infty} r^*(u) \gamma(u+s\tau) du. \text{ We call } r^*(u) = r(u)/\alpha \text{ the window of the estimator.}$$

If the variation in $\hat{\nu}$ is neglected the standard error of $\hat{\gamma}(s\tau)$ is $1/\alpha$ times the standard error of $\hat{C}_Y(s\tau)$, an estimator of which is given, e.g., in Hannan (1960).

3. Random element function

In this case an estimator of a weighted average of $\gamma(u)$ is

$$(3) \quad \hat{\gamma}(s\tau) = (\hat{C}_Y(s\tau) - \hat{\nu} c(s\tau))/\alpha; s=0, \pm 1, \pm 2, \dots,$$

where $c(v) = \int_{-\infty}^{\infty} E_X[w(u,X)w(u+v,X)] du$. It can be shown that $\hat{\gamma}(s\tau)$ is an

approximately unbiased estimator of $\int_{-\infty}^{\infty} r^*(u)\gamma(u+s\tau)du$, where

$$r(u) = \int_{-\infty}^{\infty} E_X[w(u+z,X)] E_X[w(z,X)] dz \text{ and } r^*(u) \text{ and } \alpha \text{ are related as in section 2.}$$

The standard error of $\hat{\gamma}(s\tau)$ is, again, approximately $1/\alpha$ times the standard error of $\hat{C}_Y(s\tau)$.

4. Monte Carlo experiments

4.1 Fixed element function

Experiments were carried out with two types of point processes: the Poisson and the normal point process. (For the definition of a normal point process see Linhart and Zucchini (1972).) The element functions used were $2e^{-ku}; k=5; 10; 15$. Typical results are given in Figure 1.

It is clear that the window, $r^*(u)$, depends on k , the parameter of the element function. As k increases the band-width of the window decreases and the "smudging" is less severe.

Unless the covariance density is very distinct (as, e.g., in the normal case, Fig. 1) the smudging can obliterate even its most important features.

In this respect Figure 1 should not be regarded as typical, in many cases the covariance density is much smoother and tends quickly to zero. This difficulty cannot be rectified by reducing the band-width since in practice the element function, which determines the band-width, is beyond the control of the experimenter.

To investigate the behaviour of the estimator the covariance density of the same process was estimated using 50 independent realisations of 200 observations. The results, for the Poisson and normal case, are given in Tables 1 and 2 respectively. The bias in Table 2 is due to smudging, it decreases with decreasing band-width (increasing k). Decreasing the band-width seems, however, to increase the standard error. The smudging, and thus the bias are not evident in Table 1 because the true covariance is constant in this case.

4.2 Random element function

The point processes generated were again Poisson and normal. The following element functions were used.

(a) Exponential with $w(t-T_i; X_i) = X_i e^{-5(t-T_i)}$,

where the X_i are independently and exponentially distributed.

(b) Rectangular with $w(t-T_i; X_{1i}, X_{2i}) = X_{1i}$ for $0 \leq t-T_i \leq X_{2i}$, zero otherwise,

where the X_{ji} were assumed to be independently and exponentially distributed.

The remarks made for fixed element functions concerning the smudging of important features of the covariance density apply here too.

Typical results are given in Figure 2. Here the weighting function considered is exponential. A decrease in ν , the parameter of the random amplitude brings about an increase in the band-width and hence a greater degree of "smudging".

To conclude: the usefulness of the estimators depends mostly on the shape of the weighting function $r^*(u)$. This function, or at least its band-width, has to be calculated in every instance if an estimated covariance density is to be interpreted intelligently.

We found also that the estimators are strongly influenced by $r(sr)$ in the fixed case and by $c(sr)$ in the random case. This means that if in an application the element function assumed is *wrong* (i.e. deviates from the one actually operating on the point process) the estimators can be very misleading. This seems to be particularly the case for random element functions, even if only the assumed distribution of the random vector is incorrect.

TABLE 1

Averages and standard deviations of 50 independent estimates of the covariance density, each based on 200 observations on a **filtered Poisson process** ($\lambda=1$). True covariance density is zero. Fixed element function: $2e^{-ku}$.

Averages

$\begin{matrix} x \\ k \end{matrix}$	0,0	0,5	1,0	1,5	2,0
5	-0,044	-0,003	-0,011	0,016	0,000
10	0,069	-0,048	-0,008	-0,021	-0,018
15	-0,113	-0,007	-0,026	-0,011	-0,030

Standard deviations

$\begin{matrix} x \\ k \end{matrix}$	0,0	0,5	1,0	1,5	2,0
5	0,2145	0,1000	0,0775	0,0775	0,0632
10	0,4393	0,1095	0,1304	0,1378	0,1483
15	0,6519	0,1549	0,1483	0,1612	0,1817

TABLE 2

Averages and standard deviations of 50 independent estimates of the covariance density, each based on a **filtered $N(1,0; 0,0025)$ renewal process**. Fixed element function: $2e^{-ku}$.

Averages

$\begin{matrix} x \\ k \end{matrix}$	0,0	0,5	1,0	1,5	2,0
5	-0,978	-0,779	1,088	-0,562	0,960
10	-0,984	-0,983	2,543	-0,939	2,096
15	-1,041	-0,996	3,430	-0,988	2,752
True Values	-1,000	-1,000	6,980	-1,000	2,990

Standard deviations

$\begin{matrix} x \\ k \end{matrix}$	0,0	0,5	1,0	1,5	2,0
5	0,0316	0,0000	0,0447	0,0000	0,0447
10	0,1761	0,0447	0,1643	0,0447	0,1612
15	0,3130	0,0548	0,3302	0,0548	0,3332

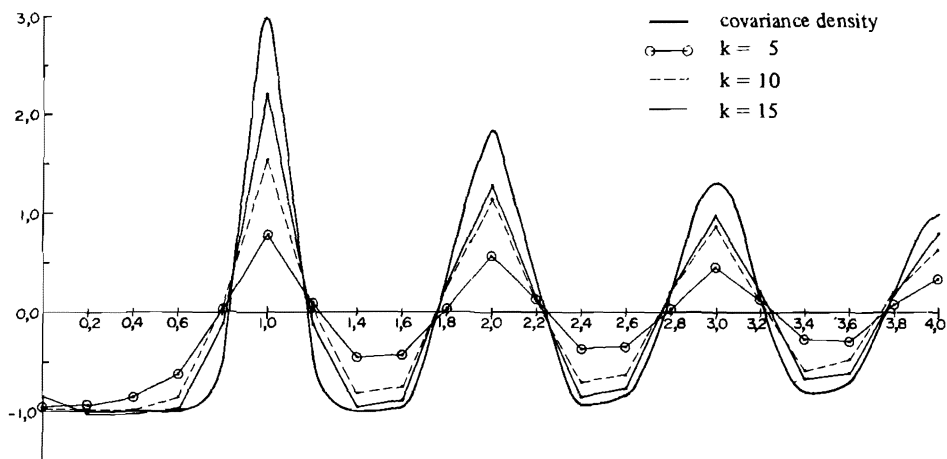


FIGURE 1: $N(1,0; 0,01)$ renewal process, $n = 10\,000$, $\tau = 0,1$. Fixed element function $2e^{-ku}$

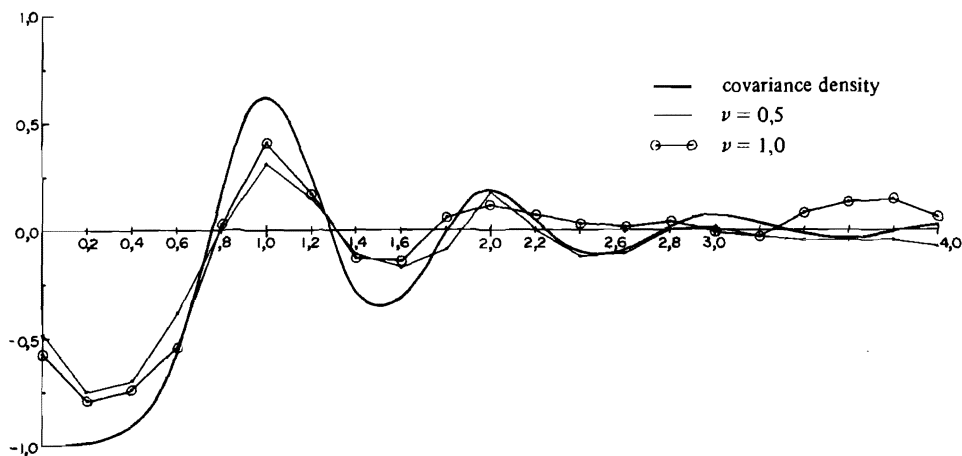


FIGURE 2: $N(1,0; 0,0625)$ renewal process, $n = 10\,000$, $\tau = 0,1$. Random element function Xe^{-5u} where $X \sim \text{Exp}(\nu)$.

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